

SW Team - 3rd Party Module Guidelines

Page Content

- [General](#)
- [Copyright Header](#)
- [Changelog](#)
- [Open Source Licenses](#)



SEOS-2326 - Getting issue details...

STATUS

TODO:

- Clarify location of this page, include in / align with [GPL Tracker](#) pages?
- Explain or define external repo structure, branch names to avoid collisions?
- Enforce 3rd party folder structure more strictly (e.g. no 3rd party folder currently for TimeServer, NIC_iMX6)?
- File and copyright header: The approach below is used for a fork/copy-case. Explain different approach upstream-case?
- Follow the guidelines from <https://reuse.software/tutorial> and <https://spdx.dev/ids/>?
- There is a paragraph focused on handling licenses (GPL's copy left issue).
- Check if we have/want guidelines for SPDX license identifier for our own open source files.

LIMITATIONS:

- LICENSE.md does not directly point to the 3rd party module location (e.g. modules using mbedtls just mention mbedtls but do not point to libs/3rdParty/mbedtls).
- If HC files are added to a 3rd party folder as a change, the [LICENSE.md](#) / license-text for those files is only found in the module using the 3rd party module at the moment (e.g. [LICENSE.md](#) in os_network_stack for picotcp).

General

- In general **only required changes** shall be made to 3rd party code (e.g. no coding style adaptations), unless the module is completely reworked.
(WHY: Support easy comparison and allow upstreaming.)
- If possible, the 3rd party modules shall be clearly **separated** from the files created by HC.
(WHY: Have a clear module structure.)
- Templates are included in the doc_templates repo: https://bitbucket.hensoldt-cyber.systems/projects/HC/repos/doc_templates.

Example 3rd Party Folder Structure

```
NIC_RPi/  
  3rdParty/  
    uspi  
      ...  
      CHANGELOG.md  
      ...  
    ...  
    LICENSE.BSD-3-Clause  
    LICENSE.md  
    ...
```

Copyright Header

Definitions:

- Copyright header: Header comment including copyright (and licensing) information.

- File header: Header comment including information about the module.

1. The **HC copyright header** should be added to **all added and changed files** within the 3rd party module.

(WHY: Easy to follow rule compared to decisions on a per file bases.)

- A separate copyright header shall be added above the 3rd party file/copyright header instead of including the information in the original file header.
(WHY: Consistent approach for different 3rd party modules. Readability despite long 3rd party file/copyright headers.)
- 3rd party copyright lines may be included in the HC copyright header (ascending order) if there is no 3rd party file/copyright header or it is replaced completely.
(WHY: If 3rd party modules are reworked it might make sense to replace the 3rd party file/copyright header.)
- Only a short file description may be included in the HC copyright header. Otherwise a separate (doxygen) file header should be placed below the HC copyright header.
(WHY: Pragmatic approach to avoid two very short headers for copyright and file.)

2. The year in the copyright header shall be the **year the change was implemented**.

- If substantial changes have been done in several years, a **year range <first year>-<last year>** should be used.
- If in doubt whether a change is substantial, prefer the year range.

3. If applicable, the appropriate **SPDX license identifier** (cf. <https://spdx.org/licenses/>) for the 3rd party module shall be added.

Example HC Copyright Header Above 3rd Party Copyright Header

```
/*
 * Copyright (C) 2020, HENSOLDT Cyber GmbH
 * SPDX-License-Identifier: BSD-2-Clause
 */

/*
 * Copyright 2017, Data61
 * Commonwealth Scientific and Industrial Research Organisation (CSIRO)
 * ABN 41 687 119 230.
 *
 * This software may be distributed and modified according to the terms of
 * the BSD 2-Clause license. Note that NO WARRANTY is provided.
 * See "LICENSE_BSD2.txt" for details.
 *
 * @TAG(DATA61_BSD)
 */
```

Example HC Copyright Header With 3rd Party Copyright

```
/*
 * Copyright (C) 2011-2013 Mike McCauley
 * Copyright (C) 2020-2021, HENSOLDT Cyber GmbH
 * SPDX-License-Identifier: GPL-3.0-only
 */
```

Example HC Copyright Header With Short Description

```
/*
 * Optional: Module name or short description
 *
 * Copyright (C) 2020, HENSOLDT Cyber GmbH
 * SPDX-License-Identifier: BSD-2-Clause
 */
```

Example HC Copyright Header Above File Header

```
/*
 * Copyright (C) 2020, HENSOLDT Cyber GmbH
 * SPDX-License-Identifier: BSD-2-Clause
 */

/**
 * @file
 * @brief Module name or short description
 * ...
 */
```

Changelog

1. For every 3rd party module a changelog shall contain the **notable changes** made by HC within the module.
(WHY: Give overview about relevant changes.)
 - Changes shall be described briefly on a level required for a user to understand the consequences and reasoning behind.
 - As for commit message, imperative present tense (i.e. ""Fix bug" and not "Fixed bug" or "Fixes bug") should be used to describe changes.
 - The commit history should not be duplicated, no detailed change history or the listing of all changed files is required.
(WHY: 3rd party module can be compared with the baseline version and the commit history is available internally or upon request.)
2. The file should be called **CHANGELOG.md** within the 3rd party folder.
 - If there is already a CHANGELOG.md in the 3rd party folder, the changes might be added to the existing file but the changes by HC should be distinguishable from the original changes.
3. The changelog shall have the following structure:
 - The **baseline version of the 3rd party module** shall be mentioned, including the source/link if possible.
(WHY: Support easy comparison.)
 - Also any updates of the baseline version later on shall be mentioned.
 - **Changes between release versions should be grouped together.**
(WHY: Changes for the user are only relevant between releases.)
 - If **no notable changes** are made for a new release version, no section for this version should be created.
(WHY: No need to update the module if there are no changes.)
 - New changes may be grouped below ## [UnreLeased] if the release version is not known yet.
(WHY: Support changelog creation during development.)
 - Same types of changes should be grouped below the categories:
 - **Added,**
 - **Removed,**
 - **Changed,**
 - **Fixed,**
 - **Security,**
 - **Deprecated.**
4. The changelog format shall consider:
 - Only **basic markdown features** (headings, lists, code) shall be used.
(WHY: Keep file readable in text-only.)
 - The line width should not exceed **80 characters**.
(WHY: Keep file readable in text-only.)

(Refer to <https://keepachangelog.com/en/1.0.0/> for a basic changelog concept.)

Example CHANGELOG.md

Changelog

All notable changes by HENSOLDT Cyber GmbH to this 3rd party module included in the TRENTOS SDK will be documented in this file.

For more details it is recommended to compare the 3rd party module at hand with the previous versions of the TRENTOS SDK or the baseline version.

[1.1]

Security

- Fix vulnerability ...

[1.0]

Fixed

- Fix wrong calculation of ...

Added

- Update XYZ library to version 1.3.0.

[0.9]

Changed

- Rename error codes ...

Added

- Start development based on XYZ library 1.2.3 (<<https://github.com/abc/xyz>>).

Open Source Licenses

1. License files within the 3rd party folder shall be preserved.
2. A file **LICENSE.md** shall be added to a module using an Open Source module (e.g. parent-folder).
 - a. The license for files created by HC shall be mentioned.
 - b. The used Open Source modules shall be listed with their respective license and source.
3. The Open Source license text for files created by HC shall be added in a file called **LICENSE.<SPDX-License-Identifier>** (e.g. LICENSE.BSD-3-Clause).

Example LICENSE.md

License

All files in this module are released under BSD-3 license, except those from the 3rd party modules mentioned below.

The BSD-3 license can be found in: LICENSE.BSD-3-Clause

3rd Party Modules

The table lists the 3rd party modules used within this module, their licenses and the source from which they were obtained:

Name	SPDX Identifier	Source
bcm2837	GPL-3.0-or-later	< https://www.airspayce.com/mikem/bcm2835/index.html >
spiflash	MIT	< https://github.com/pellepl/spiflash_driver >

